Memory Notes

Simplified DDR2 Simplified edition - JEDEC JESD79-2F

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Table of Contents

1 Simplified Ddr2 Index....10
2 Introduction....11
2.1 What DDR2 SDRAM is....11
2.2 Key features of DDR2....12
2.3 Densities and speed grades....13
2.4 The simplified drill model....14
3 Organization....15
3.1 Banks, rows, columns....15
3.2 Addressing....16
3.3 Prefetch and IO width....17
3.4 Ranks and stacked packages....18
3.5 The drill model restated....19
4 Acronyms and Terms....20
5 Power-Up and Initialization....23
5.1 Cold start sequence....24
5.2 Notes that bite....25
5.3 Why the order matters....26
6 Mode Registers....27
6.1 MR - Mode Register (BA1=0, BA0=0)....28
6.2 EMR1 - Extended Mode Register 1 (BA1=0, BA0=1)....29
6.3 EMR2 - Extended Mode Register 2 (BA1=1, BA0=0)....30
6.4 EMR3 - Extended Mode Register 3 (BA1=1, BA0=1)....31

6.5 Programming rules....32
7 Command Truth Tables....33
7.1 Command truth table....34
7.2 CKE truth table (synchronous transitions)....37
7.3 Data mask truth table....38
7.4 Not in the table....39
8 Commands: Rules and Constraints....40
8.1 ACT - Bank Activate....41
8.2 RD - Read (A10=0)....42
8.3 WR - Write (A10=0)....43
8.4 PRE / PREA - Precharge....44
8.5 REF - Refresh....45
8.6 Self-refresh entry / exit....46
8.7 Power-down....47
8.8 NOP and DES....48
8.9 Interrupts....49
9 Walkthrough 1: AXI4 Read, Row Hit....50
9.1 Controller configuration used in all Chapter 4 walkthroughs....50
9.2 Scenario....51
9.3 Command trace....52
9.4 What happens on the AXI side....53
9.5 Scheduling takeaways....54
10 Walkthrough 2: AXI4 Read, Row Miss....55
10.1 Scenario....56

10.2 Command trace....57
10.3 Cost accounting....58
10.4 AXI side....59
10.5 Variation: read with auto precharge (RDA)....60
11 Walkthrough 3: AXI4 Write to an Idle Bank....61
11.1 Scenario....62
11.2 Command trace....63
11.3 AXI side....64
11.4 Write recovery is the trap....65
11.5 Comparison against the read miss....66
12 Walkthrough 4: Mixed Traffic and Bus Turnarounds....67
12.1 Scenario....68
12.2 Command trace....69
12.3 The two turnaround rules....70
12.4 Scheduling takeaways....71
12.5 AXI side....72
13 Data Path: Prefetch, Bursts, Strobes....73
13.1 4n prefetch....73
13.2 Burst length and burst order....74
13.3 DQS: the source-synchronous strobe....75
13.4 Write data mask (DM)....76
13.5 Latency arithmetic, collected....77
14 ODT - On-Die Termination....78
14.1 What it is....78

14.2 Programming.....79
14.3 ODT timing.....80
14.4 System rules.....81
14.5 What DDR2 does NOT have.....82
15 Core Timing Parameters.....83
15.1 Row-cycle parameters.....84
15.2 Activate-rate parameters (cross-bank).....85
15.3 Data-to-precharge parameters.....86
15.4 Why these four numbers dominate.....87
16 Turnaround and Burst Timing.....88
16.1 CAS-to-CAS and latencies.....89
16.2 Direction changes.....90
16.3 Strobe-level timing (for completeness).....91
16.4 Burst interrupt rules.....92
17 Refresh and Self-Refresh Timing.....93
17.1 Refresh parameters.....93
17.2 Self-refresh.....94
17.3 Power-down exit timings (related CKE rules).....95
18 Package and Electrical Notes.....96
18.1 Packages.....97
18.2 Supplies and signaling.....98
18.3 Speed bins.....99
18.4 Capacitance (loading budget).....100
19 Test and Calibration Features.....101

19.1 OCD - Off-Chip Driver impedance adjustment.....102
19.2 Test mode (MR A7).....104
19.3 What is not here.....105

List of Figures

No figures in this document.

List of Tables

No tables in this document.

List of Waveforms

No waveforms in this document.

1 Simplified Ddr2 Index

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2 Introduction

2.1 What DDR2 SDRAM is

DDR2 SDRAM is the second generation of double-data-rate synchronous DRAM, standardized by JEDEC as JESD79-2F. It transfers two data words per clock cycle per pin (one on each edge of the data strobe), and it quadruples the internal prefetch relative to single-data-rate DRAM: the array is accessed four words at a time (4n prefetch) so the IO can run at twice the data rate of DDR while the core runs at roughly half the IO rate.

Family positioning, in one line each:

Tech	Prefetch	Distinctives
SDR SDRAM	1n	Single data rate, one word per clock
DDR SDRAM	2n	Double data rate, 2.5 V SSTL_2
DDR2 SDRAM	4n	ODT, posted CAS / additive latency, 1.8 V SSTL_1.8
DDR3 SDRAM	8n	Bank groups arrive later (DDR4), fly-by topology

2.2 Key features of DDR2

- 4n prefetch architecture: every column access moves 4 words internally.
- Burst length 4 or 8 only. There is no burst-chop and no Burst Terminate command; BL4 bursts can never be interrupted.
- Posted CAS with additive latency (AL): a Read or Write command may be issued immediately after Activate and is held inside the DRAM for AL clocks, so the command bus is freed early. Read latency RL = AL + CL, write latency WL = RL - 1.
- On-die termination (ODT): programmable termination resistance (off, 75, 150, or 50 ohm nominal) on the DQ/DQS/DM pins, controlled by an ODT pin.
- Off-chip driver (OCD) impedance calibration: the DRAM can trim its own output driver strength against a system reference.
- Differential data strobes (DQS/DQS#), source-synchronous in both directions.
- DLL (delay-locked loop) aligns DQS to the external clock for reads; it is enabled at initialization and reset under software control.
- No bank groups (those are a DDR4 concept). Banks are a flat set of 4 or 8.
- No DBI, no parity, no on-die ECC. Data integrity is entirely the controller's problem.
- Supply: VDD = VDDQ = 1.8 V (SSTL_1.8), VREF = VDDQ/2, VTT termination rail at 0.9 V nominal.

2.3 Densities and speed grades

Standard monolithic densities run from 128 Mb to 4 Gb, in x4, x8 and x16 organizations. Speed grades are named by data rate: DDR2-400, DDR2-533, DDR2-667 and DDR2-800, each with one or more CL-tRCD-tRP bin combinations (e.g. DDR2-800E is the 6-6-6 bin).

2.4 The simplified drill model

Real parts have thousands of rows and over a thousand columns per row. All examples in this book instead use a deliberately tiny geometry so bank state is obvious at a glance:

- Banks: B0-B7 (eight, as on 1 Gb and larger parts)
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

This model is used by the ddr_drills training app. Every command trace, walkthrough and timing example in Chapters 3, 4 and 6 uses it. Real address maps are far larger; the scheduling rules are identical.

Source: JESD79-2F sections 1, 3.2, 3.6.1

3 Organization

3.1 Banks, rows, columns

A DDR2 device is a set of independent banks. Each bank is a 2D array addressed by row then column. Activating a bank copies one row into that bank's sense amplifiers (the "open row" or page); reads and writes then hit the sense amps at column granularity. A bank must be precharged (closed) before a different row in the same bank can be opened.

Density	Banks	Bank address	Notes
128 Mb	4	BA0-BA1	x16 only
256 Mb	4	BA0-BA1	
512 Mb	4	BA0-BA1	
1 Gb	8	BA0-BA2	
2 Gb	8	BA0-BA2	
4 Gb	8	BA0-BA2	

There are no bank groups in DDR2. All banks share one timing regime: tRRD applies between any two Activates, tCCD between any two column commands, and tFAW limits any rolling window of four Activates on 8-bank parts.

3.2 Addressing

The row address rides with the Activate command on A0-A15 (width depends on density). The column address rides with the Read or Write command; A10 is not a column bit - it is the auto-precharge flag (often written AP). Column addressing is A0-A9, plus A11 on x4 organizations, so a page holds 1024 or 2048 words.

Page size formula: page size = 2^COLBITS x ORG / 8 bytes, per bank.

Density	Organization	Row bits	Column bits	Page size
1 Gb	x8	A0-A13	A0-A9	1 KB
1 Gb	x16	A0-A12	A0-A9	2 KB
2 Gb	x8	A0-A14	A0-A9	1 KB
4 Gb	x16	A0-A14	A0-A9	2 KB

3.3 Prefetch and IO width

The 4n prefetch means one column access pulls 4 words per DQ pin group from the array in parallel. Consequences:

- Minimum burst is 4 data words per pin (BL4); BL8 takes two internal 4-word fetches.
- The array core runs at one quarter of the pin data rate.
- Internal read-to-precharge and write-recovery analog delays are defined from the last 4-word prefetch, which is why BL4 and BL8 sometimes get different formulas for the same parameter (tRTP is the classic case).

IO organizations are x4, x8 and x16, each with one DQS pair per 8 bits (x4 shares one strobe per 4 bits).

3.4 Ranks and stacked packages

The spec also covers dual-die and quad-die stacked packages. A stack presents multiple ranks behind one channel: each rank gets its own chip select (and its own ODT control), shares the address/command and data busses, and is otherwise a complete independent DDR2 device. Rank-to-rank traffic adds bus-turnaround constraints beyond single-rank rules.

3.5 The drill model restated

This book's examples use 8 banks (B0-B7), 8 rows per bank (R0-R7) and 8 columns per row (C0-C7). One bank's state at a glance:

Bank	State	Open row
B0	idle	-
B3	active	R5
B5	active	R1

Source: JESD79-2F sections 2.2, 2.4, 3.2

4 Acronyms and Terms

Terms used with precise meanings throughout this book.

Term	Meaning
ACT	Bank Activate command: opens a row in a bank
AL	Additive latency: clocks a posted CAS is held inside the DRAM (EMR1)
AP	Auto precharge flag, carried on address bit A10 with RD/WR
BA	Bank address pins (BA0-BA2)
BL	Burst length: 4 or 8 data words per column command
CAS	Column Address Strobe (command pin); also shorthand for a column command
CKE	Clock enable; gates power-down and self-refresh entry/exit
CL	CAS latency: read-command-to-data delay after the posted hold (MR)
DCC	Duty Cycle Corrector (optional, EMR2)
DES	Deselect command (CS high)
DLL	Delay-locked loop aligning read DQS to CK
DM	Data mask pin (write data masking, per byte)
DQS	Data strobe; source-synchronous clock for DQ in both directions
EMR(n)	Extended Mode Register n (1, 2, 3)
MRS / EMRS	Mode Register Set / Extended Mode Register Set commands
MR	Mode Register (burst, CL, WR, DLL reset, PD-exit mode)
NOP	No Operation command

Term	Meaning
OCD	Off-Chip Driver impedance calibration
ODT	On-Die Termination (Rtt on DQ/DQS/DM, per EMR1)
PASR	Partial Array Self Refresh (EMR2)
PD	Power-down (precharge PD or active PD)
PRE	Precharge command (A10=0: one bank; A10=1: all banks)
RD / WR	Read / Write commands (A10=0)
RDA / WRA	Read / Write with auto precharge (A10=1)
RDQS	Read data strobe (optional, replaces DM on x8 parts)
REF	Refresh command (auto refresh, one internal row per bank)
RL	Read latency = AL + CL
Rtt	ODT termination resistance value
SRE / SRX	Self-refresh entry / exit
SSTL_1.8	1.8 V stub-series-terminated logic IO standard
tCK	Clock period
WL	Write latency = RL - 1
Bank group	A DDR4 concept; does not exist in DDR2
Page / open row	The row currently held in a bank's sense amplifiers

Command trace notation used in Chapters 3 and 4:

```
ACT B3 R5      # open row R5 in bank B3
RD  B3 C2      # read burst starting at column C2
WR  B1 C0      # write burst starting at column C0
RDA B2 C4      # read with auto precharge
PRE B3         # precharge bank B3
PREA           # precharge all banks
```

```
REF             # auto refresh
--- (tWTR bubble) ---   # forced idle bus time, reason named
```

Source: JESD79-2F sections 2.3, 3, 4.1

5 Power-Up and Initialization

DDR2 must be initialized by a fixed sequence; anything else is undefined behavior. The sequence below paraphrases the required steps.

5.1 Cold start sequence

1. Ramp supplies with CKE held below 0.2 x VDDQ and the ODT pin LOW. VDD/VDDL and VDDQ ramp together (or VDD first), VTT last; each ramp must complete within spec windows (200 ms VDD, 500 ms VDDQ-then-VTT). VREF tracks VDDQ/2 and must never exceed VDDQ.
2. Start the clock (CK/CK#) and keep it stable.
3. Wait at least 200 us with stable power and clock, issuing NOP or Deselect, then raise CKE.
4. Wait at least 400 ns more (NOP/Deselect), then issue Precharge-All.
5. EMRS to EMR(2) - BA1=1, BA0=0 selects it.
6. EMRS to EMR(3) - BA1=1, BA0=1.
7. EMRS to EMR(1) with A0=0: enable the DLL.
8. MRS with A8=1: DLL reset (this also programs the other MR fields).
9. Precharge-All.
10. Two or more Refresh commands.
11. MRS with A8=0: program operating parameters without resetting the DLL.
12. At least 200 clocks after step 8, run OCD calibration. If OCD is not used, instead write EMR(1) twice: once with the OCD-default code (A9=A8=A7=1), once with the OCD-exit code (A9=A8=A7=0), carrying the desired EMR(1) operating fields.
13. Ready for normal operation.

The required gaps between these commands (tRP, tMRD, tRFC) look like this:

```
... CKE high ... 400 ns ... PREA ... EMRS(2) ... EMRS(3) ...
EMRS(1) ...
     MRS(DLL reset) ... PREA ... REF ... REF ... MRS(operating) ...
     >= 200 clk ... OCD flow ... ready
        |tRP|tMRD|tMRD|tMRD|   |tRP|tRFC|tRFC|tMRD|
```

5.2 Notes that bite

- All four mode registers must be written; the spec allows them in any order but the canonical sequence above programs EMR2/EMR3 first because their fields are inert defaults.
- The register contents are undefined at power-up - there is no hardware default. Skip a register and its fields are garbage.
- MRS/EMRS require all banks precharged and idle, CKE high, and tMRD (2 clocks) before the next command.
- Reprogramming MR/EMR later is legal any time all banks are idle, and does not disturb array contents - but any field not rewritten takes its newly written value, so every field must always be written, even unchanged ones.
- The DLL needs 200 clocks after enable+reset before a Read may be issued, or tAC/tDQSCK are not guaranteed. This same 200-clock rule reappears at self-refresh exit (tXSRD).
- DLL is automatically disabled in self-refresh and re-enabled on exit.

5.3 Why the order matters

DLL enable (step 7) must precede DLL reset (step 8); the reset pulse is only meaningful on an enabled DLL. The precharge-all before refresh guarantees a clean idle state; the two refreshes seed the internal refresh counter so the first tREFI window is well-defined. OCD calibration must follow all mode-register programming because its results depend on the configured drive strength and ODT settings.

Source: JESD79-2F sections 3.3, 3.3.1

6 Mode Registers

Four registers, selected by bank address during an MRS/EMRS command: MR (BA=00), EMR1 (BA=01), EMR2 (BA=10), EMR3 (BA=11). All must be programmed at init; all writes require all banks idle and tMRD spacing.

6.1 MR - Mode Register (BA1=0, BA0=0)

Field	Bits	Meaning
Burst length	A2-A0	010 = BL4, 011 = BL8 (only two legal values)
Burst type	A3	0 = sequential, 1 = interleaved
CAS latency	A6-A4	CL 2-6 by encoding; no half-clock latencies in DDR2
Test mode	A7	Must be 0 for normal operation
DLL reset	A8	1 = issue DLL reset (self-clearing)
Write recovery	A11-A9	WR in clocks: codes for 2, 3, 4, 5, 6
PD exit mode	A12	0 = fast exit (tXARD), 1 = slow exit (tXARDS)

WR is programmed in clocks, computed by rounding tWR[ns] / tCK up. It also feeds tDAL (write auto-precharge recovery + precharge) as WR + tRP.

6.2 EMR1 - Extended Mode Register 1 (BA1=0, BA0=1)

Field	Bits	Meaning
DLL enable	A0	0 = enable (required for normal operation)
Output drive	A1	0 = full strength, 1 = reduced strength
Rtt (ODT)	A6, A2	00 = off, 01 = 75 ohm, 10 = 150 ohm, 11 = 50 ohm
Additive latency	A5-A3	AL = 0, 1, 2, 3, 4 (5 optional)
OCD program	A9-A7	000 exit, 001 drive(1), 010 drive(0), 100 adjust, 111 default
DQS disable	A10	1 = single-ended DQS mode (DQS# tied off externally)
RDQS enable	A11	1 = DM pin becomes RDQS (x8 parts; disables DM)
Qoff	A12	1 = output buffers disabled (IDD measurement aid)

Notes: the 50 ohm Rtt option is mandatory at DDR2-800, optional below. If RDQS is enabled, the write data mask function is lost on that pin. When issuing later EMR1 writes for other reasons, A9-A7 must be 000 or the OCD setting is disturbed.

6.3 EMR2 - Extended Mode Register 2 (BA1=1, BA0=0)

Refresh-related features; everything else reserved-zero.

Field	Bits	Meaning
PASR	A2-A0	Partial array self refresh: full, half, quarter, 1/8, 3/4 array (which banks depends on code; optional feature)
DCC	A3	Duty Cycle Corrector enable (optional, may not be controllable)
SRF	A7	High-temperature self-refresh rate enable (>85 C; optional)

If PASR is set beyond full array, data outside the selected banks is lost on self-refresh entry. If unsupported, PASR must be 000.

6.4 EMR3 - Extended Mode Register 3 (BA1=1, BA0=1)

No function is defined. All address bits reserved-zero. It must still be written during initialization.

6.5 Programming rules

- Precondition: all banks precharged, CKE high.
- Every MRS/EMRS write redefines the whole register; always drive every field intentionally.
- tMRD = 2 clocks between any MRS/EMRS and the next command.
- Reserved bits and BA2/A13-A15: drive 0.

Source: JESD79-2F sections 3.4, 3.4.1, 3.4.2

7 Command Truth Tables

All commands are registered on the rising edge of CK from the state of CS, RAS, CAS, WE (and CKE across two cycles for power modes). H = high, L = low, X = either (but a valid level). BA selects the bank, or the target register for MRS/EMRS.

7.1 Command truth table

Command	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	Address
MRS / EMRS	H	H	L	L	L	L	OP	BA=reg, op code on A
Refresh (REF)	H	H	L	L	L	H	X	X
Self-refresh entry	H	L	L	L	L	H	X	X
Self-refresh exit	L	H	H/X or L	X/H	X/H	X/H	X	X
Precharge one bank	H	H	L	L	H	L	L	BA = bank
Precharge all (PREA)	H	H	L	L	H	L	H	X
Activate (ACT)	H	H	L	L	H	H	row	BA = bank, row on A
Write (WR)	H	H	L	H	L	L	L	BA, column

Command	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	Address
Write w/ auto pre (WRA)	H	H	L	H	L	L	H	BA, column
Read (RD)	H	H	L	H	L	H	L	BA, column
Read w/ auto pre (RDA)	H	H	L	H	L	H	H	BA, column
NOP	H	X	L	H	H	H	X	X
Deselect (DES)	H	X	H	X	X	X	X	X
Power-down entry	H	L	H/X or L	X/H	X/H	X/H	X	X
Power-down exit	L	H	H/X or L	X/H	X/H	X/H	X	X

Mnemonics:

- ACT = RAS-only (RAS low, CAS and WE high).
- PRE = RAS + WE (CAS high); A10 picks one bank vs all banks.
- RD = CAS-only (WE high); WR = CAS + WE; A10 on either means “with auto precharge”.
- MRS = all three strobes low; BA picks which of the four registers.

- REF = RAS + CAS low, WE high; same encoding with CKE falling is self-refresh entry instead.
- Power-down entry is any NOP/DES encoding with CKE falling; exit is CKE rising with NOP/DES.

7.2 CKE truth table (synchronous transitions)

Current state	CKE prev	CKE cur	Command	Result
Power-down	L	L	X	stay in power-down
Power-down	L	H	DES or NOP	exit power-down
Self-refresh	L	L	X	stay in self-refresh
Self-refresh	L	H	DES or NOP	exit self-refresh
Bank(s) active	H	L	DES or NOP	active power-down entry
All banks idle	H	L	DES or NOP	precharge power-down entry
All banks idle	H	L	REF	self-refresh entry

7.3 Data mask truth table

DM is sampled with write data, one per byte lane: DM = 0 writes the byte, DM = 1 masks it (the array keeps its old contents). There is no read data masking. If RDQS is enabled (x8 parts), the pin is RDQS during reads and DM is unavailable.

7.4 Not in the table

No Burst Terminate command exists in DDR2 (DDR had one; DDR2 dropped it). BL4 bursts are uninterruptible; BL8 bursts may only be interrupted by a same-direction burst aligned to the 4-word boundary.

Source: JESD79-2F sections 4.1, 4.2, 4.3, 3.6.2

8 Commands: Rules and Constraints

Each command, what it does, and what it forbids next. Timing symbols are defined in Chapter 6; the drill model (B0-B7, R0-R7, C0-C7) is used for examples.

8.1 ACT - Bank Activate

Opens row R in bank B (row copied to the bank's sense amps). Required before any RD/WR to that bank.

- ACT -> RD/WR same bank: wait tRCD, unless additive latency absorbs it. With AL > 0 the RD/WR may be issued immediately ("posted") and the DRAM holds it for AL clocks; AL must still be chosen so internal CAS lands no earlier than tRCD.
- ACT -> ACT different bank: tRRD.
- ACT -> ACT same bank: tRC (= tRAS + tRP), and the bank must have been precharged first.
- On 8-bank parts: at most 4 ACTs in any rolling tFAW window.
- ACT -> PRE same bank: tRAS (row must stay open long enough to restore).

8.2 RD - Read (A10=0)

Burst read of BL words starting at the given column. Data appears RL = AL + CL clocks after the command; DQS goes low one clock before data (read preamble).

- RD -> RD (any bank): tCCD = 2 clocks; seamless back-to-back every BL/2 clocks.
- RD -> WR: read-to-write turnaround, BL/2 + 2 clocks command spacing (4 clocks at BL4, 6 at BL8), so the write preamble does not collide with the read burst.
- RD -> PRE same bank: earliest at AL + BL/2 + max(RTP, 2) - 2 clocks; also bounded by the analog tRTP from the last 4-word prefetch, and by tRAS from the ACT.
- RDA (A10=1): the bank auto-precharges; the precharge begins (AL + BL/2) clocks after the command, delayed internally until tRAS and tRTP are met. Next ACT to that bank: tRP after precharge actually starts, and tRC from the old ACT.

8.3 WR - Write (A10=0)

Burst write; first data is strobed in WL = RL - 1 clocks after the command. Write data must respect tDQSS (DQS aligned to CK).

- WR -> WR (any bank): tCCD = 2 clocks; seamless every BL/2 clocks.
- WR -> RD: CL - 1 + BL/2 + tWTR clocks after the WR command. tWTR covers moving the last write data from the input buffer into the sense amps - it is not the same thing as write recovery.
- WR -> PRE same bank: WL + BL/2 + tWR. tWR (write recovery) lets the last data be written into the array before the row closes. tWR is programmed in clocks via MR; violating it corrupts data.
- WRA (A10=1): auto precharge starts after write recovery; the bank may be re-activated once (WR + tRP) - the tDAL combination - and tRC are met.

8.4 PRE / PREA - Precharge

Closes the open row, restoring data to the array. A10=0 closes the bank selected by BA; A10=1 closes all banks.

- PRE -> ACT same bank: tRP. On 8-bank parts, PREA gets one extra clock: tRPall = tRP + 1 tCK.
- Illegal while a burst it would truncate is still in flight (see RD/WR rules above).

8.5 REF - Refresh

One internal refresh step across all banks; address counter is internal.

- Precondition: all banks precharged and tRP met.
- REF -> ACT or REF: tRFC (density-dependent, 75-327.5 ns).
- Average rate tREFI = 7.8 us (3.9 us above 85 C case, optional mode).
- Scheduling slack: up to 8 REFs may be postponed (“posted”), so the worst gap between two REFs is 9 x tREFI.

8.6 Self-refresh entry / exit

Entry: REF encoding with CKE falling, all banks idle, ODT off. The DRAM keeps itself refreshed with the external clock ignored; only CKE matters.

- Minimum stay: tCKE (3 clocks).
- Exit: clock must be stable first, then CKE high. NOP/DES during the exit window. tXSNR (= tRFC + 10 ns) before any non-read command; tXSRD (200 clocks) before a read - the DLL is re-locking.
- Before re-entering self-refresh, issue at least one REF (a refresh may have been missed at the exit boundary).
- PASR (EMR2) limits refresh to a fraction of the array; data outside it is lost on entry.

8.7 Power-down

CKE low with NOP/DES. Two flavors: precharge PD (all banks idle; DLL off) and active PD (a row open). Active PD has two exit modes selected by MR A12: fast exit (tXARD = 2 clocks, DLL kept on) and slow exit (tXARDS = 6-AL clocks, lower power, DLL off).

- Exit to any non-read command: tXP = 2 clocks.
- CKE low pulse must be at least tCKE.
- PD performs no refresh: duration is bounded by refresh requirements, 9 x tREFI at best with 8 REFs posted beforehand.

8.8 NOP and DES

Bus fillers. NOP keeps CS low (command bus active); DES raises CS. Neither changes state. Used to occupy forced gaps (tRRD, tCCD, turnaround bubbles).

8.9 Interrupts

BL4: no interruption of any kind. BL8: a read may be interrupted by a read, a write by a write, only at the 4-word boundary. Cross-direction or misaligned interruption is illegal.

Source: JESD79-2F sections 3.5-3.11, 3.15, 3.16

9 Walkthrough 1: AXI4 Read, Row Hit

9.1 Controller configuration used in all Chapter 4 walkthroughs

Drill model geometry (B0-B7, R0-R7, C0-C7), BL4, posted CAS enabled:

Parameter	Value (clocks)
CL / AL / RL / WL	3 / 2 / 5 / 4
tRCD / tRP / tRAS / tRC	3 / 3 / 6 / 9
tRRD / tFAW / tCCD	2 / 10 / 2
tWTR / tWR / tRTP	3 / 6 / 3

The AXI4 data bus is assumed as wide as one DRAM burst, so one AXI4 beat maps to one DRAM burst word: an AXI4 burst of 4 beats is exactly one BL4 DRAM burst. Real controllers slice and dice; the drill model does not.

9.2 Scenario

Initial bank state: B3 has row R5 open. All other banks idle.

AXI4 master issues: ARADDR mapping to bank B3, row R5, column C2, ARLEN=3 (4 beats), ARSIZE matching full bus width.

9.3 Command trace

```
RD  B3 C2        # row R5 already open in B3 - no ACT needed, go
straight to CAS
```

One command. That is the whole point of a row hit.

Timeline:

```
clk:      0        1    2    3    4    5    6    7
CMD:      RD
DQS:                        pre  ----DQS toggling----
DQ:                              D0   D1   D2   D3
```

Data appears RL = AL + CL = 5 clocks after the RD command (AL=2 was programmed, but with no ACT in front there is nothing to post against; the DRAM still applies RL from the RD command itself).

9.4 What happens on the AXI side

- AR handshake completes; the controller's address mapper splits the address into {B3, R5, C2}.
- The bank machine for B3 reports R5 open -> issue RD immediately, skipping ACT entirely.
- At RL the DRAM drives DQS/DQ; the PHY returns 4 data words; the controller streams them out as 4 AXI R beats with RLAST on the final word.

9.5 Scheduling takeaways

- Row-hit reads are the cheapest DRAM transaction: 1 command, latency RL.
- The bank stays open afterwards (A10=0). A follow-up access to R5 is another hit; an access to a different row of B3 now costs a PRE + ACT (see Walkthrough 2).
- A controller doing FR-FCFS scheduling reorders the queue exactly to manufacture this case.

Source: JESD79-2F sections 3.5, 3.6.1, 3.6.3

10 Walkthrough 2: AXI4 Read, Row Miss

Same controller configuration as Walkthrough 1 (CL=3, AL=2, RL=5, WL=4, BL4; tRCD=3, tRP=3, tRAS=6).

10.1 Scenario

Initial bank state: B5 has row R6 open (from earlier traffic). AXI4 read arrives: ARADDR maps to bank B5, row R2, column C1, ARLEN=3.

Row R2 is not the open row -> row miss. The bank must be closed and the right row opened before the read.

10.2 Command trace

```
PRE B5           # close R6 - wrong row is open in B5, precharge
before re-activating
--- (tRP = 3) ---# sense amps need tRP to finish restoring before the
next ACT
ACT B5 R2        # open the requested row R2
RD  B5 C1        # posted CAS: issued 1 clk after ACT, DRAM holds it
AL=2 clks;
                 # internal CAS lands at ACT+3, exactly tRCD
```

Timeline:

```
clk:      0     1    2    3     4     5     6    7    8    9   10   11
CMD:      PRE                    ACT   RD
                |<-- tRP=3 -->|       |AL=2|
                                      |<------ tRCD=3 (internal) ------>|
DQ:
D0 ... D3
```

- PRE at clk 0, ACT legal at clk 3 (tRP).
- RD posted at clk 4 (one after ACT); additive latency holds it to clk 4 + 2 = clk 6 internally, satisfying tRCD = 3 from ACT at clk 3.
- Read data starts at clk 4 + RL = clk 9.

10.3 Cost accounting

Row miss penalty = tRP + tRCD of pure dead time on this bank before data can flow: 6 clocks here, versus 0 for a hit. Posted CAS hides the command bus occupancy (the RD command slot is spent early, at clk 4), but it does not hide the latency - internal CAS still cannot beat tRCD.

10.4 AXI side

- The AR channel accepts the request; the controller cannot produce R beats until the DRAM data comes back, so the master just sees a longer RVALID latency. Nothing on the AXI interface changes - all of this is hidden inside the controller.
- If the master had a second read outstanding to a different (idle) bank, a good scheduler would run that one first during the tRP/tRCD bubble.

10.5 Variation: read with auto precharge (RDA)

If the controller predicts no further use of R2 soon, it issues `RDA B5 C1` (A10=1) instead. The bank then precharges itself (AL + BL/2 = 4 clocks after the RD command, delayed if tRAS or tRTP is not yet met) and is idle for the next tenant without a separate PRE command slot.

Source: JESD79-2F sections 3.5, 3.6.1, 3.6.3, 3.7.1, 3.8.1

11 Walkthrough 3: AXI4 Write to an Idle Bank

Same controller configuration (CL=3, AL=2, RL=5, WL=4, BL4; tRCD=3, tRP=3, tWR=6).

11.1 Scenario

Initial state: all banks idle. AXI4 write arrives: AWADDR maps to bank B1, row R4, column C0, AWLEN=3, followed by 4 W-channel beats.

An idle bank needs no precharge - miss-to-idle is cheaper than miss-to-open-row by tRP.

11.2 Command trace

```
ACT B1 R4         # open the row; bank was idle so no PRE needed
WR  B1 C0         # posted CAS, 1 clk after ACT; internal CAS at ACT+3
= tRCD
--- (WL = 4) --- # write latency: first DQS edge lands 4 clks after
the WR command
--- (tWR = 6) ---# write recovery: last data must soak into the array
before PRE
PRE B1            # close the row (only if the controller wants the
bank back)
```

Timeline:

```
clk:      0     1     2    3    4    5    6    7    8    9   ...  13
CMD:      ACT   WR
                          |<---- WL=4 ---->|
DQS:                                pre -DQS toggling-
DQ:                                  D0   D1   D2   D3
                                                        |<- tWR=6 ->|
PRE legal at clk 4 + 4(WL) + 2(BL/2) + 6(tWR) = clk 16 from ACT,
i.e. WL + BL/2 + tWR = 12 clocks after the WR command.
```

11.3 AXI side

- AW and W are independent channels; the controller buffers the 4 W beats (or streams them) while the DRAM side runs ACT + posted WR.
- BREADY goes back to the master once the controller has accepted responsibility for the data - typically after the write is safely in flight to the DRAM, not after tWR completes.
- Write data is strobed in with DQS driven by the controller/PHY; tDQSS alignment to CK is the PHY's job.

11.4 Write recovery is the trap

tWR is not a turnaround nicety; it is array physics. Precharging B1 before tWR expires destroys the data just written - the bits never finish committing from the sense amps to the cells. This is why WR is a mode-register field: the DRAM itself uses the programmed WR value to time auto precharge after a WRA, and tDAL = WR + tRP is the real “write to next activate” cost when auto precharge is used.

11.5 Comparison against the read miss

	Read miss (walkthrough 2)	Write to idle bank
Commands before data	PRE, ACT, RD	ACT, WR
Dead time before data	tRP + tRCD	tRCD (hidden by WL buffering)
Tail cost	none (bank left open)	tWR before PRE

Source: JESD79-2F sections 3.5, 3.6.1, 3.6.4, 3.7.2, 3.8.2

12 Walkthrough 4: Mixed Traffic and Bus Turnarounds

Same controller configuration (CL=3, AL=2, RL=5, WL=4, BL4; tWTR=3, tWR=6, tCCD=2, tRRD=2, tFAW=10).

The shared DQ bus can only face one way at a time. Read-to-write and write-to-read direction changes cost idle bus clocks - the turnaround bubbles - and they are where bandwidth goes to die in mixed traffic.

12.1 Scenario

Four AXI4 requests, in arrival order:

1. Read B3 R5 C2 (B3 has R5 open - hit)
2. Write B1 R4 C0 (B1 idle)
3. Read B2 R1 C6 (B2 idle)
4. Read B3 R5 C7 (same open row as #1 - hit)

12.2 Command trace

```
RD  B3 C2        # hit: data flows immediately, RL=5
--- (RD->WR: BL/2+2 = 4 clk command gap) ---   # bus must clear the
read burst
ACT B1 R4        # schedule during the turnaround bubble - command bus
is free
WR  B1 C0        # posted; first DIN at WR+WL(4); bus turned around
--- (WR->RD: CL-1+BL/2+tWTR = 7 clk) ---       # last DIN must reach
the array
ACT B2 R1        # again hidden inside the bubble
RD  B2 C6        # next read legal 7 clks after the WR
RD  B3 C7        # hit; tCCD=2 after previous RD - seamless
```

Timeline (clock numbers approximate the spacing rules):

```
clk:    0    1    2    3    4    5    6    7    8    9   10   11   12
13   14
CMD:    RD                          WR
RD   RD
             |<- 4 ->|                 |<------ 7 ------>|
bus:    .... D0-D3 .. ..    p  D0-D3 .. .. .. ..    p  D0-D3 D0-D3
dir:        RD->   (bubble)  WR->      (bubble)   RD->    RD->
```

12.3 The two turnaround rules

Transition	Command spacing	Why
RD -> WR	BL/2 + 2 (4 clk at BL4)	Read burst + postamble must clear the bus before the write preamble
WR -> RD	CL - 1 + BL/2 + tWTR	Last write data must travel from the input buffer into the sense amps before a read reuses them

tWTR is internal and invisible on the bus, but the command spacing rule enforces it. Note it does NOT include WL: the spacing is expressed from the WR command, and CL - 1 already accounts for WL = RL - 1 = AL + CL - 1 with AL cancelled against the posted-CAS hold.

12.4 Scheduling takeaways

- Turnarounds are more expensive than same-direction tCCD gaps: 4 and 7 clocks here versus 2. Grouping reads with reads and writes with writes (write batching / read prioritization) is the single biggest bandwidth lever in mixed traffic.
- ACT commands to other banks are nearly free: the command bus is idle during data turnarounds, so row misses on other banks hide completely.
- tFAW still applies: this trace's two ACTs are 9 clocks apart, fine; a burst of five ACTs within 10 clocks would stall the fifth.
- With only 8 banks in the drill model, tFAW (4 ACTs per 10 clocks) can bind before tRRD does on an activate-heavy stream - watch both.

12.5 AXI side

The AXI fabric never sees any of this. Read masters see variable RVALID latency; write masters see variable WREADY/BVALID timing. A controller with enough queue depth reorders requests 1-4 (respecting AXI ordering rules per ID) to convert the trace above into two reads back-to-back and the write parked until the bus is free.

Source: JESD79-2F sections 3.5, 3.6.3, 3.6.4

13 Data Path: Prefetch, Bursts, Strobes

13.1 4n prefetch

Every column access moves 4 words per IO group between the array and the IO buffers at once. The array core therefore runs at one quarter of the pin data rate, which is the fundamental trick that let DDR2 double DDR's bus speed without a proportionally faster core.

Consequences that show up everywhere else in the spec:

- BL4 is the minimum burst; BL8 takes two consecutive 4-word internal fetches.
- Analog delays tied to data movement (tRTP, tWTR, tWR) are defined from the last 4-word prefetch, not from the command - which is why BL4 and BL8 versions of the same rule differ by 2 clocks.
- Burst interrupts, where legal at all, are only legal on 4-word boundaries (the prefetch granularity).

13.2 Burst length and burst order

BL is 4 or 8, set in MR. Burst type (MR A3) picks the address sequence within the burst:

BL	Start A1A0	Sequential	Interleaved
4	00	0,1,2,3	0,1,2,3
4	01	1,2,3,0	1,0,3,2
4	10	2,3,0,1	2,3,0,1
4	11	3,0,1,2	3,2,1,0

BL8 follows the same wrap-and-pair-swap pattern over A2-A0 (sequential wraps in 4-word nibbles; interleaved swaps adjacent pairs within each nibble). Sequential mode at BL8 is nibble-based: the low 4 words wrap within their aligned 4-word group before the upper 4.

The practical use of interleaved order: a critical-word-first cache-line fill, where the requested word arrives first and the rest of the line follows in pair-swapped order.

13.3 DQS: the source-synchronous strobe

DQ data is not referenced to CK at the pin; it is referenced to DQS.

- Reads: the DRAM drives DQS edge-aligned with DQ. DQS drops low one clock before the first data (read preamble, tRPRE ~ 0.9-1.1 tCK) and returns to tri-state after a postamble (tRPST). tDQSCK bounds DQS against CK (DLL-controlled), tDQSQ bounds DQ skew against DQS, tQH the hold.
- Writes: the controller drives DQS roughly center-aligned to each DQ window. DQS must arrive aligned to CK within tDQSS (+/- 0.25 tCK); setup and hold of DQ against DQS are tDS/tDH. Write preamble tWPRE and postamble tWPST frame the burst.
- DQS is normally differential (DQS/DQS#). Single-ended mode (EMR1 A10) exists; the complement pin must then be tied off externally.

13.4 Write data mask (DM)

One DM pin per byte lane, sampled with write data: high masks that byte (the cell keeps its old value). No read masking. On x8 parts, enabling RDQS (EMR1 A11) repurposes the DM pin as a read strobe echo - masking is lost, so partial writes become read-modify-write at the controller.

13.5 Latency arithmetic, collected

Quantity	Formula	Programmed by
RL (read latency)	AL + CL	EMR1 (AL), MR (CL)
WL (write latency)	RL - 1	derived
RD cmd -> data	RL clocks	
WR cmd -> first DQS	WL clocks	
RD -> RD, WR -> WR	max(tCCD=2, BL/2)	

Source: JESD79-2F sections 3.6, 3.6.1-3.6.5

14 ODT - On-Die Termination

14.1 What it is

DDR2 moves bus termination onto the die. A programmable termination resistor (Rtt) sits on every DQ, DQS, DQS#, DM, RDQS and RDQS# pin, and a dedicated ODT pin switches it in or out. Before ODT, termination lived on the motherboard and could not follow bus ownership; with ODT, the device that is listening terminates, and the device that is driving does not.

14.2 Programming

EMR1 bits A6 and A2 select the nominal value:

A6 A2	Rtt (nominal)
0 0	ODT disabled
0 1	75 ohm
1 0	150 ohm
1 1	50 ohm (mandatory support at DDR2-800, optional below)

The ODT pin then enables the programmed value dynamically, cycle by cycle.

14.3 ODT timing

Symbol	Meaning	Value
tAOND	ODT pin high -> termination starts turning on	2 clocks
tAON	termination actually in regulation	tAC(min) to tAC(max)+1 ns window
tAOFD	ODT pin low -> turn-off begins	2.5 clocks
tAOF	termination released	tAC(min) to tAC(max)+0.6 ns
tAONPD / tAOFPD	same, in power-down (slower)	up to ~2-2.5 tCK + tAC
tANPD	ODT low before power-down entry	3 clocks
tAXPD	ODT may be re-raised after PD exit	8 clocks
tMOD	EMRS (Rtt change) -> new value live	0-12 ns

Changing Rtt by EMRS has a protocol to avoid impedance glitches on the channel: ODT pin must be low tAOFD before the EMRS, stay low through the whole tMOD window, and only then be raised again.

14.4 System rules

- Reads: the controller side terminates (its own ODT or board Rtt); the DRAM's ODT is off while it drives.
- Writes: the target DRAM's ODT is on; in multi-rank systems the non-target rank usually terminates instead (rank-to-rank ODT steering), so the driver sees a clean load at the far end.
- ODT must be off before self-refresh entry and during the tXSRD exit window; the ODT pin is simply held low.
- ODT is not available at all during self-refresh.

14.5 What DDR2 does NOT have

Stated plainly, because later generations add all of these and it is easy to back-port them mentally:

- No DBI (data bus inversion). Every DQ bit drives its true value every cycle; there is no inversion flag pin and no power-saving inversion logic.
- No write or command parity. A corrupted command or write is silently executed.
- No on-die ECC and no ECC storage mode. System-level ECC, if wanted, is entirely the controller's job with extra DRAM devices for check bits.
- No CRC on the data bus (that arrives with DDR4's write CRC).

If your design needs any of these properties, it must build them above the DRAM, in the controller or the interconnect.

Source: JESD79-2F sections 3.4.4, 3.4.5, 3.10

15 Core Timing Parameters

The bank-state timings: what each parameter constrains, and the command transition it applies to. Values are the spec's speed-bin table entries; "1KB/2KB page" variants depend on device organization. Clock conversion: divide by tCK and round up (RU), unless the spec defines the parameter directly in clocks.

15.1 Row-cycle parameters

Symbol	Definition	Applies between	Min (typ bins)
tRCD	Row-to-column delay: time for the row to reach the sense amps before a column command may execute	ACT(bank x) -> RD/WR(bank x)	10-15 ns (800C/800D/800E); posted CAS still must land internal CAS at >= tRCD
tRP	Precharge recovery: sense amps and bitlines restored before re-activation	PRE(bank x) -> ACT(bank x)	10-15 ns, bin-matched to tRCD; PREA on 8-bank parts: tRP + 1 tCK
tRAS	Minimum row-open time: data must be restored to the cells before precharge	ACT(bank x) -> PRE(bank x)	45 ns min (40 ns DDR2-400B), 70 us max
tRC	Row cycle: full open-close-reopen loop for one bank	ACT(bank x) -> ACT(bank x)	tRAS + tRP; 55-60 ns at DDR2-800

15.2 Activate-rate parameters (cross-bank)

Symbol	Definition	Applies between	Min
tRRD	Activate-to-activate spacing, any two banks (limits peak array current)	ACT(bank x) -> ACT(bank y)	7.5 ns (1KB page) / 10 ns (2KB page)
tFAW	Four-activate rolling window: at most 4 ACTs inside any window of this length (8-bank parts)	any 4 ACTs within a sliding window	37.5 ns (1KB) / 50 ns (2KB) at 400/533/667; 35/45 ns at DDR2-800

tFAW example with the drill model: if RU(tFAW/tCK) = 10 and an ACT lands at clock N, at most three more ACTs may land in clocks N+1 .. N+9.

15.3 Data-to-precharge parameters

Symbol	Definition	Applies between	Min
tRTP	Read-to-precharge: last 4-word read prefetch must complete internally before the row closes	RD(bank x) -> PRE(bank x)	7.5 ns
tWR	Write recovery: last write data must be committed to the cells before precharge	end of WR burst(bank x) -> PRE(bank x)	15 ns; programmed in clocks in MR, WR = RU(tWR/tCK), codes for 2-6
tDAL	Write auto-precharge total: programmed WR plus precharge, before the bank may be re-activated	WRA(bank x) -> ACT(bank x)	WR + tRP (clocks)

Command-spacing formulas that fall out of these (same bank):

Transition	Earliest spacing (clocks)
RD -> PRE	AL + BL/2 + max(RU(tRTP), 2) - 2
WR -> PRE	WL + BL/2 + RU(tWR)
RDA -> ACT	AL + RU((tRTP + tRP)/tCK) at BL4; +2 more at BL8; also >= tRC
WRA -> ACT	tDAL, also >= tRC

The -2 in RD->PRE confuses everyone once: the spacing is measured from the RD command, and the analog tRTP clock starts at the internal read (AL clocks in), so the posted-CAS hold and the burst itself cover part of it. For BL8 the internal read starts 2 clocks later relative to the burst end, hence max(RTP, 2) and the BL-dependent forms.

15.4 Why these four numbers dominate

tRCD and tRP are the row-miss penalty. tRAS + tRP = tRC bounds how fast one bank can cycle rows. tRRD and tFAW bound how fast the whole device can open rows regardless of how many banks are free - on activate-heavy streams (many small transfers, open-page policy) tFAW is usually the binding constraint, not tRRD. The controller's scheduler exists to hide tRCD/tRP behind other banks' data and to pace ACTs against tRRD/tFAW.

Source: JESD79-2F sections 3.5, 3.7, 3.8, Tables 41-43

16 Turnaround and Burst Timing

Column-command spacing and bus-direction rules. These are the parameters that decide how full the DQ bus can be kept.

16.1 CAS-to-CAS and latencies

Symbol	Definition	Applies between	Value
tCCD	Column command to column command, same or different banks	RD/WR -> RD/WR (any direction same-type)	2 clocks
RL	Read latency: command to first data	RD -> first DQS/DQ out	AL + CL clocks
WL	Write latency: command to first strobe	WR -> first DQS in	RL - 1 clocks
CL	CAS latency component of RL	posted internal CAS -> data	2-6 clocks (MR)
AL	Additive latency: posted-CAS hold	RD/WR command -> internal CAS	0-4 clocks, 5 optional (EMR1)

BL/2 clocks of bus occupancy per burst: BL4 occupies 2 clocks of DQ, BL8 occupies 4. Seamless same-direction traffic means issuing a command every BL/2 clocks - at BL4 that is exactly tCCD = 2, so back-to-back BL4 bursts saturate the bus.

16.2 Direction changes

Note on tRTW: JESD79-2F specifies the read-to-write spacing directly as a relation (BL/2 + 2 clocks, section 3.6.3) without assigning it a symbol. This book calls it tRTW because the other specifications in this series (LPDDR3/4, DDR3/4/5, HBM) name that parameter, and a common symbol set keeps the timing tables comparable across books.

Symbol	Definition	Applies between	Value
tRTW	Read-to-write turnaround: read burst and postamble must clear before write preamble (book symbol; the spec gives the relation unnamed)	RD(any bank) -> WR(any bank)	BL/2 + 2 clocks: 4 at BL4, 6 at BL8
tWTR	Write-to-read: last write data must propagate from input buffer into the sense amps	end of WR burst(any) -> RD(any)	7.5 ns (10 ns at DDR2-400)

Command-spacing forms:

Transition	Spacing from first command (clocks)
RD -> WR	BL/2 + 2
WR -> RD	CL - 1 + BL/2 + RU(tWTR)
RD -> RD	max(tCCD, BL/2)
WR -> WR	max(tCCD, BL/2)

Worked example (the Chapter 4 configuration: CL=3, AL=2, RL=5, WL=4, BL4, tWTR=3 clk): RD->WR = 4 clocks, WR->RD = 2+2+3 = 7 clocks. The asymmetry is the write path's longer internal tail; reads turn into writes cheaper than writes into reads.

16.3 Strobe-level timing (for completeness)

These bound the DQS/DQ relationship rather than the scheduler, but the controller's PHY must meet them:

Symbol	Definition	Value
tDQSS	Write DQS latching edge to CK alignment	+/- 0.25 tCK
tWPRE / tWPST	Write DQS preamble / postamble	0.35 min / 0.4-0.6 tCK
tRPRE / tRPST	Read DQS preamble / postamble	0.9-1.1 / 0.4-0.6 tCK
tDQSCK	Read DQS output access from CK (DLL window)	+/- 350-500 ps by bin
tDQSQ	DQ skew against DQS	200-350 ps by bin
tQH	DQ/DQS output hold from DQS	tHP - tQHS
tDS / tDH	Write DQ setup / hold against DQS	bin-dependent, 50-275 ps base

16.4 Burst interrupt rules

- BL4: never interruptible, in either direction, by anything.
- BL8: a read may be cut by a later read, or a write by a later write, at the 4-word (prefetch) boundary only. The new command's data follows seamlessly.
- There is no Burst Terminate command. A burst always runs its full length unless legally interrupted as above.

Source: JESD79-2F sections 3.6.1-3.6.4, Tables 42-43

17 Refresh and Self-Refresh Timing

17.1 Refresh parameters

DRAM cells leak; every row must be refreshed within the retention window. Refresh is all-banks and internally addressed - the controller only schedules it.

Symbol	Definition	Applies between	Value
tREFI	Average interval between refresh commands	REF -> REF (average)	7.8 us at 0-85 C case; 3.9 us at 85-95 C (optional extended range)
tRFC	Refresh cycle time: refresh in progress, no other command	REF -> ACT, REF -> REF	75 ns (256 Mb), 105 (512 Mb), 127.5 (1 Gb), 195 (2 Gb), 327.5 (4 Gb)

Rules:

- All banks must be precharged (and tRP met) before REF.
- Up to 8 refresh commands may be postponed (posted), so the worst-case gap between two REFs is 9 x tREFI. Posting buys scheduling freedom to finish a burst sequence or a tFAW window first; it is not free - the postponed REFs land as a cluster later.
- Violating refresh timing silently corrupts data; the spec says the data must be rewritten before any valid read.
- tRFC grows steeply with density because one REF must touch more array. At 4 Gb, refresh duty cycle = tRFC/tREFI is ~4% of all bus time.

17.2 Self-refresh

The DRAM refreshes itself with the external clock stopped: for sleep states where the controller is powered down but memory contents must survive.

Symbol	Definition	Applies between	Value
tCKE	Minimum CKE pulse, high or low - sets the minimum self-refresh stay	SRE -> SRX (and any CKE pulse)	3 clocks
tXSNR	Exit self-refresh to any non-read command	SRX (CKE high) -> e.g. ACT, PRE, MRS	tRFC + 10 ns
tXSRD	Exit self-refresh to a read command	SRX -> RD	200 clocks

Entry checklist: all banks idle, ODT off (low), then REF encoding with CKE falling. The DLL shuts down automatically.

Exit checklist: stable clock first, then CKE high; NOP/DES through the exit window; wait tXSNR for non-read commands or tXSRD for reads (the DLL is re-locking - same 200-clock rule as init); issue at least one REF before re-entering self-refresh (an internally timed refresh may have been skipped at the exit boundary); keep ODT off until tXSRD is met.

PASR (EMR2 A2-A0) limits self-refresh to a fraction of the array (full, half, quarter, 1/8, 3/4 - the bank mapping depends on the code and bank count). Anything outside the refreshed region is gone on entry. The high-temperature bit (EMR2 A7) doubles the internal refresh rate above 85 C where leakage doubles.

17.3 Power-down exit timings (related CKE rules)

Symbol	Definition	Value
tXP	Precharge power-down exit to any command	2 clocks
tXARD	Active power-down exit (fast mode, DLL on) to read	2 clocks
tXARDS	Active power-down exit (slow mode, DLL off) to read	6 - AL clocks (7 - AL at 667, 8 - AL at 800)
tANPD / tAXPD	ODT low before PD entry / ODT valid after PD exit	3 / 8 clocks

Fast vs slow exit is an MR (A12) choice per system: fast keeps the DLL burning through active power-down for a 2-clock resume; slow saves that power but pays the longer exit.

Source: JESD79-2F sections 3.9, 3.10, 3.11, Table 40, Tables 42-43

18 Package and Electrical Notes

A light pass: what matters when connecting a DDR2 device, deliberately skipping the full ballout tables (consult the spec figures for those).

18.1 Packages

- Single-die parts use FBGA per MO-207; x4/x8 in a 60-ball grid, x16 in an 84-ball grid.
- Dual-die and quad-die stacked parts use MO-242; the stack presents 2 or 4 ranks with separate chip selects and per-rank ODT associations.
- Support (mechanical) balls exist in some variations - they are not signals.

18.2 Supplies and signaling

Rail	Nominal	Feeds
VDD	1.8 V	core
VDDQ	1.8 V	output drivers (isolated from VDD noise by design intent)
VDDL	1.8 V	DLL
VREF	VDDQ / 2	input comparator reference (SSTL_1.8)
VTT	0.9 V nominal	bus termination rail (board side)

- SSTL_1.8: inputs are compared against VREF; tracking between VREF and VDDQ/2 must hold within 300 mV during ramp and tightly in operation.
- Address/command and data busses are source-terminated point-to-point or short-stub topologies with VTT pull-ups at the far end (plus ODT on the data group).
- CK/CK# is differential and must be kept crossing cleanly; tCH/tCL duty window is 0.45-0.55 tCK (0.48-0.52 on the tighter 667/800 bins).
- The spec allows spread-spectrum clocking within a bounded down-spread band; the DLL tracks it within limits.

18.3 Speed bins

Grade	tCK range (CL bin)	Typical CL-tRCD-tRP
DDR2-400	5.0-8.0 ns	3-3-3, 4-4-4
DDR2-533	3.75-8.0 ns	3-3-3, 4-4-4
DDR2-667	3.0-8.0 ns	4-4-4, 5-5-5
DDR2-800	2.5-8.0 ns	4-4-4, 5-5-5, 6-6-6

All bins share the same core analog floors (tRCD/tRP/tRAS scale with the bin numbers in ns), which is why higher-speed parts need more clocks for the same nanoseconds - and why a controller should store timings in ns and convert to clocks at configuration time.

18.4 Capacitance (loading budget)

Ballpark from the spec tables: CK inputs 1-2 pF, other inputs 1-2 pF, DQ/DQS group 2.5-4 pF. These set how many devices a CK/address tree can drive before a register or buffer is required - the reason registered DIMMs exist.

Source: JESD79-2F sections 2.1-2.3, 6 (Tables 17, 39, 41)

19 Test and Calibration Features

DDR2 has almost no test logic compared to later generations. What it has is OCD calibration, a test-mode bit you must never set, and no boundary scan at all.

19.1 OCD - Off-Chip Driver impedance adjustment

The output driver strength is trimmable so the DQ/DQS drive impedance can be matched to the channel instead of relying on process-lot luck. Control lives in EMR1 A9-A7:

A9 A8 A7	Operation
0 0 0	Exit OCD calibration mode; keep current setting
0 0 1	Drive(1): DQ/DQS driven high, DQS# low (measure pull-up)
0 1 0	Drive(0): DQ/DQS driven low, DQS# high (measure pull-down)
1 0 0	Adjust mode: shift the driver code (BL4 op-code on all DQs: increment, decrement, or hold)
1 1 1	OCD calibration default (nominal 18 ohm full-strength drive)

Flow:

```
for pull-up and pull-down sides:
    EMRS Drive(1) or Drive(0)      # drive a static level
    measure at the controller/PHY
    EMRS OCD exit                  # mandatory after every calibration
command
    if adjustment needed:
        EMRS Adjust mode           # then a BL4 write of the adjust
op-code
        EMRS OCD exit
EMRS OCD default (if not calibrating) then EMRS OCD exit
```

Rules:

- Every OCD calibration command must be followed by an OCD-exit EMRS before any other command.
- All mode registers must be programmed before OCD calibration, and ODT must be managed so it does not corrupt the measurement.
- OCD applies only to the full-strength drive setting; with reduced strength (EMR1 A1) the OCD characteristics do not apply, and the default values are likewise not applicable when Adjust mode is used.

- If OCD is not used at all, the init sequence still requires the default-then-exit pair (see Chapter 2).
- tOIT (0-12 ns) covers the output delay after entering a drive mode.

19.2 Test mode (MR A7)

MR A7 selects a vendor test mode. The spec's only directive for users is: keep it 0. There is no standardized user-visible test mode.

19.3 What is not here

- No boundary scan / JTAG on the DRAM itself (that lives on registers and buffers in module form factors, not in the JESD79-2 device spec).
- No connectivity test mode, no per-pin loopback, no MPR (multi-purpose register) reads - those arrive with DDR3.
- No post-package repair interface exposed to the user.
- The DLL is observable only indirectly, through read-data timing (tAC/tDQSCK); there is no lock-status readback.

Source: JESD79-2F sections 3.4.1, 3.4.3